

Process Characterization Report

A-Mab Drug Substance — Ultrafiltration / Diafiltration (Step 10) — PCR-010

Process Development / Manufacturing Science & Technology (MSAT)

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Table of contents

Approvals	3
Abbreviations	3
Executive summary	5
1 Introduction	6
1.1 Product and unit operation	6
1.2 Objectives	6
1.3 Regulatory and scientific basis	6
2 Prior knowledge and quality risk basis	7
2.1 Platform and prior-product knowledge	7
2.2 Quality attributes and process-performance measures	7
2.3 Risk-based prioritization and parameter selection	7
3 Materials and methods	9
3.1 Scale-down model and its qualification	9
3.2 Operation	9
3.3 Analytical methods	9
3.4 Sampling plan	10
3.5 Statistical methods	10
4 Study design	11
4.1 Parameters, ranges and the knowledge space	11
4.2 Univariate ranging	11
5 Results	12
5.1 Process performance and reproducibility	12
5.2 Univariate ranging of the operating parameters	12
5.3 No product-quality impact	12
6 Process performance and consistency	13
7 Parameter classification	14
8 Contribution to the control strategy	15
9 Discussion	16
10 Conclusions	17
Deviations from the plan	18

References	19
Appendix A — Univariate ranging summary	20
Appendix B — Analytical methods summary	21

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Formulation / Drug Product Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Analytical Development	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

BLA, Biologics License Application · CPP, critical process parameter · CQA, critical quality attribute · DF, diafiltration · DP, drug product · DS, drug substance · DV, diavolume · GPP, general process parameter · ICH, International Council for Harmonisation · KPP, key process

parameter · MSAT, Manufacturing Science and Technology · NOR, normal operating range ·
PAR, proven acceptable range · PPQ, process performance qualification · QbD, Quality by Design
· TFF, tangential-flow filtration · TMP, transmembrane pressure · UF, ultrafiltration · UF/DF,
ultrafiltration / diafiltration.

Executive summary

This report documents the Stage 1 (Process Design) evaluation of the A-Mab final ultrafiltration / diafiltration (UF/DF) operation (Step 10), the tangential-flow-filtration step that concentrates the virus-filtration pool and buffer-exchanges it into the final formulation buffer to deliver the drug substance at its target concentration. In the A-Mab case study this operation has **no drug-substance product-quality impact**: it is a formulation / mass-balance operation, and the **formulation characterization** (excipient levels, pH, osmolality and stability) is conducted and reported under the drug-product development program rather than in this drug-substance characterization campaign. The operation was therefore evaluated on a qualified bench-scale UF/DF model against its process-performance measures, following the lifecycle approach to process validation (U.S. Food and Drug Administration 2011; Parenteral Drug Association 2013; International Society for Pharmaceutical Engineering 2023) and the Quality-by-Design principles of ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023, 2012), with the process model of the A-Mab case study (CMC Biotech Working Group 2009).

3 process parameters were evaluated one factor at a time. The evaluation confirms that UF/DF **forms or modifies no drug-substance product-quality CQA** and that it robustly concentrates and buffer-exchanges the pool to the drug-substance target.

Key outcomes

- No drug-substance product-quality CQA is set or modified by UF/DF; the glycan, charge-variant, aggregate and impurity attributes are established and cleared upstream and pass through the operation unchanged. Formulation attributes are characterized under the drug-product program and are out of scope here.
- The UF/DF step yield is **97.1%** at the nominal condition (56,232 g product mass in, 54,588 g out), completing the purification train at an overall process yield of **83.2%** and delivering **54.6 kg** of drug substance.
- The operation delivers the drug substance at its target concentration of **75 g/L** (final-concentration NOR 70–80 g/L), with diafiltration completing the exchange into the final formulation buffer.
- The number of diavolumes, the transmembrane pressure and the final DS concentration are all classified **KPP** (they govern buffer-exchange completeness, process flux/time and the final concentration — process performance — without a DS CQA impact). No parameter is a CPP.

The UF/DF operation is demonstrated to be well understood and robust as a formulation / mass-balance step. Because it forms no drug-substance product-quality attribute, its parameter ranges and control are justified on process-performance grounds, and this report rolls up into the Process Characterization Master Report (**PCMR-001**).

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose product-quality attributes — Fc N-glycosylation, charge variants, aggregate and the process-related impurities — are established in the production bioreactor and controlled (formed and cleared) through the purification train (CMC Biotech Working Group 2009). Ultrafiltration / diafiltration is the final drug-substance operation: a tangential-flow-filtration step in which ultrafiltration concentrates the virus-filtration pool and diafiltration exchanges it into the final formulation buffer, delivering the drug substance at its target concentration. The operation removes water and exchanges buffer; it does not form, remove or modify any drug-substance product-quality attribute, and is therefore evaluated here as a process-performance / mass-balance operation. The formulation characterization proper — excipient levels, pH, osmolality and stability — is conducted and reported under the drug-product development program.

1.2 Objectives

The objectives of the evaluation were to (i) confirm that UF/DF has no impact on the drug-substance product-quality CQAs; (ii) establish the effect of the operating parameters on the process-performance measures — step yield, buffer-exchange completeness and the final DS concentration; (iii) confirm that the drug substance is delivered at its target concentration and buffer composition across the operating ranges; (iv) establish proven acceptable ranges (PARs) and normal operating ranges (NORs) for each parameter; and (v) classify each parameter and provide the ranges that justify the Stage 2 operating conditions and the control strategy.

1.3 Regulatory and scientific basis

The work follows the lifecycle approach to process validation of the FDA 2011 guidance (U.S. Food and Drug Administration 2011), PDA Technical Report 60 (Parenteral Drug Association 2013) and the ISPE Good Practice Guide (International Society for Pharmaceutical Engineering 2023), and applies the enhanced (QbD) development approach of ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023, 2012). Criticality is treated as a continuum (ICH Q9). The report structure corresponds to the Stage 1 Process Design Report of PDA TR 60 §3.11 and provides development-report content suitable to support a Biologics License Application; it is correspondingly focused, because the operation forms no drug-substance CQA and the formulation characterization is reported with the drug product.

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The characterization ranges and set-points are supported by platform knowledge from related humanized IgG1 antibodies and by A-Mab development and engineering experience (CMC Biotech Working Group 2009). Prior knowledge established that the glycan, charge-variant and aggregate CQAs are set in cell culture and cleared through the purification train, and that a well-controlled UF/DF step concentrates and buffer-exchanges the pool without altering these attributes: the antibody is retained by the ultrafiltration membrane while water and small solutes permeate, so the product-quality attributes are unchanged by the operation. This knowledge framed the process-performance measures in scope and the decision to evaluate the operation univariately.

2.2 Quality attributes and process-performance measures

UF/DF **sets or modifies no drug-substance product-quality CQA**; the CQAs established upstream pass through the operation unchanged. The operation was therefore evaluated against the process-performance measures it governs (Table 2.1): the product (step) yield, the buffer-exchange completeness delivered by the diafiltration, and the final DS concentration. Formulation attributes (excipient concentrations, pH, osmolality) are drug-product characterization endpoints and are addressed under the drug-product program, not here.

Table 2.1: Process-performance measures evaluated for the UF/DF operation (no drug-substance product-quality CQA is set by this step).

Process-performance measure	Result at nominal condition	Basis / criterion
UF/DF step yield	97.1%	Mass balance across the final operation
Final DS concentration	75 g/L	Within NOR 70–80 g/L (Table 4.1)
Buffer-exchange completeness	Within target	Diavolumes / residual old-buffer target
Product-quality CQAs	Unchanged	Set and cleared upstream; not modified by UF/DF

2.3 Risk-based prioritization and parameter selection

Consistent with the A-Mab lifecycle of iterative risk assessments, the evaluation scope was set by the Pre-Characterization Process Risk Assessment (**RA-001**). Because UF/DF carries no credible

risk of impact to a drug-substance product-quality CQA, none of its parameters was assigned to a multivariate design of experiments; each parameter was evaluated one factor at a time across its characterization range for its effect on the process-performance measures. The parameters and ranges carried into this evaluation are given in Table [4.1](#).

3 Materials and methods

3.1 Scale-down model and its qualification

All runs were performed on a qualified bench-scale UF/DF (tangential-flow-filtration) model representing the commercial operation — the platform membrane chemistry and molecular-weight cut-off, operated at the commercial-equivalent membrane loading, transmembrane pressure and cross-flow, with the same ultrafiltration-concentration and diafiltration-exchange sequence (**SOP-2013**). The model was qualified against at-scale reference data (**SOP-1001**) by comparing step yield, the final concentration and the buffer-exchange completeness, confirming that the model is representative and suitable to support the parameter ranges (Parenteral Drug Association 2013).

3.2 Operation

The virus-filtration pool is concentrated by ultrafiltration, diafiltered against the final formulation buffer for the target number of diavolumes, and further concentrated to the final drug-substance target, with the transmembrane pressure and cross-flow controlled through the operation. The number of diavolumes, the transmembrane pressure and the final DS concentration are the operating variables; membrane lot and buffer composition are controlled to platform specifications and treated as identity/quality-controlled inputs rather than characterized variables (CMC Biotech Working Group 2009).

3.3 Analytical methods

Product concentration and the product-quality attributes were measured by validated methods (Table 3.1). The DS concentration was determined by protein A (**AMV-3019**); size variants (aggregate/HMW) by SEC-HPLC (**AMV-3011**) and charge variants by icIEF (**AMV-3013**) to confirm the attributes are unchanged across the operation. Method performance characteristics are summarized in Appendix B.

Table 3.1: Validated analytical methods used in the evaluation.

Reference	Title	Type
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3019	Protein Concentration by A280 (UV)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

3.4 Sampling plan

Each run was sampled at the final drug-substance pool for concentration and for the size- and charge-variant attributes, and the product mass across the operation was recorded for the step-yield calculation; the diafiltration exchange was monitored to confirm buffer-exchange completeness. Runs at the parameter set-points were replicated to quantify the reproducibility of yield and final concentration.

3.5 Statistical methods

Because no drug-substance product-quality CQA is affected, the operation was evaluated univariately rather than by a designed experiment. Each parameter was varied one factor at a time across its characterization range, holding the others at their set-points, and the process-performance measures — step yield, buffer-exchange completeness and final DS concentration — were summarized against the parameter level. Reproducibility of yield and final concentration was estimated from the replicated set-point runs. Parameter ranges were judged acceptable where the performance measures remained within their platform targets and the product-quality CQAs were unchanged.

4 Study design

4.1 Parameters, ranges and the knowledge space

3 parameters were evaluated (Table 4.1). The characterization range of each parameter (“PAR” column) defines the edges of the explored knowledge space and the basis of the proven acceptable range; the NOR is the tighter range at which the parameter is controlled routinely. Classification is an output of the evaluation (§8).

Table 4.1: UF/DF process parameters, set-points, ranges and post-characterization classification.

Parameter	Unit	Set-point	NOR	PAR	Class	Study
Number of diavolumes	DV	8	7–9	7–10	KPP	univariate
Transmembrane pressure	psi	12	10–14	10–15	KPP	univariate
Final DS concentration	g/L	75	70–80	65–85	KPP	univariate

4.2 Univariate ranging

Each parameter was evaluated one factor at a time across its characterization range; none was carried into a multivariate design because prior knowledge and the Pre-Characterization Risk Assessment indicated no credible drug-substance CQA impact. The product-quality CQAs were confirmed unchanged across the ranging.

5 Results

5.1 Process performance and reproducibility

At the nominal (set-point) condition the operation recovered antibody with a UF/DF step yield of 97.1% (56,232 g product mass in, 54,588 g out), completing the purification train at an overall process yield of 83.2% and delivering 54.6 kg of drug substance at the target concentration of 75 g/L. The set-point replicates confirmed that step yield and final concentration are reproducible, establishing that the operation delivers drug substance of consistent concentration and buffer composition.

5.2 Univariate ranging of the operating parameters

Across the characterization range of each parameter, the process-performance measures remained within their platform targets and the product-quality CQAs were unchanged:

- **Number of diavolumes.** Increasing the number of diavolumes improves the completeness of the buffer exchange (lowering residual old-buffer components) at the cost of longer process time; buffer-exchange completeness and yield remained acceptable across the range, and the NOR was confirmed as the window that assures complete exchange without unnecessary process time.
- **Transmembrane pressure.** Increasing the transmembrane pressure raises the permeate flux and shortens process time but, above the range, risks membrane fouling and concentration polarization; flux and yield remained within target across the range, and the NOR was confirmed as the window that balances throughput against fouling.
- **Final DS concentration.** As the controlled output attribute of the operation, the final DS concentration was delivered on target across the operating conditions, confirming a consistent drug-substance concentration.

5.3 No product-quality impact

The size-variant (aggregate) and charge-variant results confirmed that these attributes are unchanged across the operation, and the glycan attributes — set in cell culture — are not modified by concentration or buffer exchange. The evaluation therefore confirms that UF/DF has no drug-substance product-quality impact, consistent with the A-Mab case study, in which the operation is a formulation / mass-balance step and the formulation characterization is reported with the drug product (CMC Biotech Working Group 2009).

6 Process performance and consistency

Because the operation forms no drug-substance product-quality CQA, its robustness is expressed as the consistency of its process-performance measures rather than as a multivariate design space or a CQA capability index. The UF/DF yield of 97.1% and the delivery of drug substance at the 75 g/L target, both reproducible at the set-point and stable across the parameter ranges, establish that the operation reliably delivers drug substance of consistent concentration and buffer composition at commercial scale. The commercial-scale CQA capability is reported for the CQA-forming and CQA-clearing unit operations in their respective reports and consolidated in **PCMR-001**; UF/DF contributes to it only by preserving yield and delivering the drug substance on target.

7 Parameter classification

Each parameter was classified on the A-Mab continuum from the demonstrated effects on process performance and the absence of a drug-substance CQA impact (Table 7.1). The rationale is as follows.

- **Number of diavolumes, transmembrane pressure, final DS concentration — KPP.**
Each governs process performance — buffer-exchange completeness, permeate flux and process time, and the final concentration delivered — but none significantly impacts any drug-substance CQA; they are controlled for process performance and to deliver the drug substance on target.

No UF/DF parameter is a CPP, because the operation forms no drug-substance product-quality attribute; formulation-related criticality, where applicable, is assessed under the drug-product program.

Table 7.1: Post-characterization parameter classification.

Parameter	Class	Study
Number of diavolumes	KPP	univariate
Transmembrane pressure	KPP	univariate
Final DS concentration	KPP	univariate

8 Contribution to the control strategy

UF/DF is not a control point for any drug-substance product-quality CQA; its contribution to the overall control strategy is to complete the purification train at consistent yield and to deliver the drug substance at its target concentration and buffer composition. That contribution comprises: control of the number of diavolumes, the transmembrane pressure and the final DS concentration to their NORs for reproducible buffer exchange, throughput and concentration; in-process monitoring of the final concentration and buffer-exchange completeness; and membrane life-cycle control. The drug-substance product-quality CQAs are set and cleared upstream (reported in the paired unit-operation reports and consolidated in **PCMR-001**), and the formulation control strategy is established under the drug-product program.

9 Discussion

The evaluation confirms the expected role of UF/DF: a formulation / mass-balance operation that concentrates and buffer-exchanges the pool to the drug-substance target without forming or modifying any drug-substance product-quality attribute. The operating parameters influence buffer-exchange completeness, throughput and the final concentration in the directions established by platform knowledge, and their ranges are justified on process-performance grounds. The principal limitation is inherent to scale-down evaluation: absolute yield, flux and buffer-exchange kinetics may shift modestly at commercial scale, which is why the model is confirmed at scale during Stage 2 and the KPP ranges are carried forward with in-process monitoring of the final concentration and buffer exchange. The evaluation did not identify any parameter requiring critical (CPP) control within the studied ranges; formulation characterization is addressed under the drug-product program.

10 Conclusions

The A-Mab UF/DF operation is well understood and robust as a formulation / mass-balance step. It sets or modifies no drug-substance product-quality CQA; it recovers antibody at a step yield of 97.1%, completes the process at an overall yield of 83.2%, and delivers 54.6 kg of drug substance at the 75 g/L target; and its parameters are classified KPP with a data-supported rationale. The parameter ranges and classifications provide the basis for the Stage 2 operating ranges and the process-performance elements of the drug-substance control strategy; the formulation characterization is reported with the drug product. This report rolls up into the Process Characterization Master Report (**PCMR-001**) and provides the parameter risk basis for the post-characterization process risk assessment.

Deviations from the plan

No deviations from **PCP-010** were recorded. All planned ranging runs were executed and all planned measures were recorded on the qualified scale-down UF/DF model.

References

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- International Society for Pharmaceutical Engineering. 2023. *Good Practice Guide: Practical Implementation of the Lifecycle Approach to Process Validation*. ISPE.
- Parenteral Drug Association. 2013. *Technical Report No. 60: Process Validation — a Lifecycle Approach*. PDA.
- U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.

Appendix A — Univariate ranging summary

The parameters, characterization ranges and the process-performance measures recorded at each level. Each parameter was varied one factor at a time with the remaining parameters held at their set-points.

Table 10.1: Appendix A — one-factor-at-a-time ranging summary.

Parameter	Set-point	PAR	NOR	Class	Performance measures recorded
Number of diavolumes	8	7–10	7–9	KPP	step yield, buffer-exchange completeness, final DS concentration
Transmembrane pressure	12	10–15	10–14	KPP	step yield, buffer-exchange completeness, final DS concentration
Final DS concentration	75	65–85	70–80	KPP	step yield, buffer-exchange completeness, final DS concentration

Appendix B — Analytical methods summary

Table 10.2: Analytical methods, techniques and validation references (performance characteristics per the cited method validation reports).

Method	Technique	Analytes / measures	Validation ref.	Reportable
Protein concentration	A280 (UV)	DS concentration	AMV-3019	g/L
Size variants	SEC-HPLC	Aggregate (HMW), monomer	AMV-3011	% HMW
Charge variants	icIEF	Acidic / main / basic variants	AMV-3013	%